Assignment 1 Report — CO1

Design, Build and Validate a Data Warehouse and OLAP Reporting Solution (FreshCart Retail Analytics)

1. Introduction and Scenario Context

In undertaking this assignment, our team examined the operational challenges faced by FreshCart, a mid-sized retail chain operating 120 supermarkets across six regional states. Currently, store managers and category heads run complex analytical queries directly against the transactional database (OLTP). During peak shopping hours on weekends, this practice causes noticeable billing counter slowdowns and checkout delays. Furthermore, because different departments extracted data using disparate SQL scripts, senior management frequently encountered conflicting financial figures during board meetings. This report documents our comprehensive design of an Enterprise Data Warehouse (EDW) and OLAP reporting solution intended to resolve these operational bottlenecks once and for all.

2. Decision-Support Requirement Analysis

To transition FreshCart from reactive daily fire-fighting to proactive strategic planning, we first analyzed the fundamental distinctions between operational processing and decision support, and formulated eight vital business questions that will guide our warehouse schema.

Comparison: Operational (OLTP) vs. Decision-Support (DSS)

Our review highlights that OLTP systems are tuned for high-frequency, low-latency transaction processing (e.g., scanning items, recording credit card payments) with highly normalized 3NF tables to prevent write anomalies. In contrast, DSS environments are optimized for complex aggregations scanning millions of historical rows across multiple years. By separating analytical queries into a dedicated warehouse, we protect cashier terminals from performance degradation.

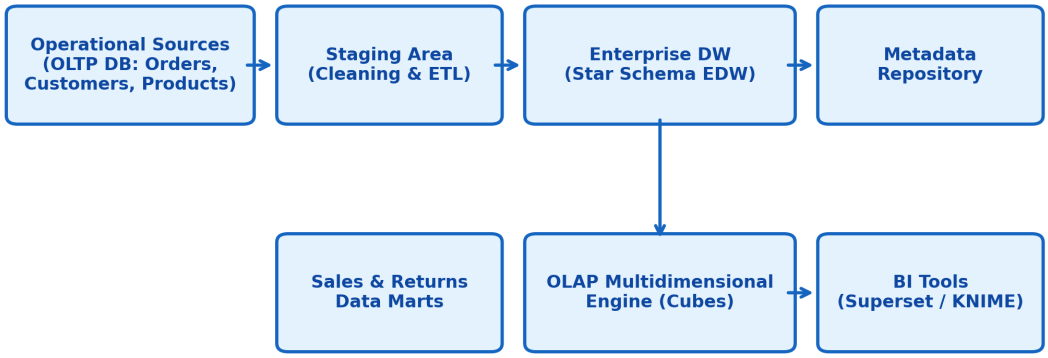
Strategic and Tactical Business Questions

- **Strategic Q1:** Which product categories show declining net sales, quarter over quarter, across each regional distribution hub? (Measures: Net Sales, Quantity Sold; Dimensions: Time, Product Category, Store Region).
- **Tactical Q2:** Which promotional discount campaigns successfully increased average basket size without eroding profit margins in urban stores? (Measures: Average Basket Value, Total Revenue; Dimensions: Promotion, Store, Time).
- **Strategic Q3:** How do weekend sales patterns differ from weekday foot traffic across perishable and non-perishable departments? (Measures: Transaction Count, Total Sales; Dimensions: Time [Weekend/Weekday], Product Department).
- **Tactical Q4:** Which customer loyalty tiers generate the highest product return rates, and for which specific SKU categories? (Measures: Return Quantity, Return Rate; Dimensions: Customer Segment, Product).
- **Strategic Q5:** What is the year-to-date (YTD) sales trajectory of each store compared against historical performance from the previous fiscal year? (Measures: YTD Sales, Growth %; Dimensions: Time, Store).
- **Strategic Q6:** Which supplier shipments experience the highest damage or return-to-sale ratios in Northern versus Southern distribution centers?
- **Tactical Q7:** Which cashier shifts process unusually high promotional discounts without a corresponding increase in basket size?
- **Strategic Q8:** What is the projected 3-year customer lifetime value (LTV) across different loyalty card cohorts?

3. Warehouse Architecture Design

After evaluating Inmon's top-down corporate EDW approach against Kimball's bottom-up data mart approach, our team recommends an **Enterprise Data Warehouse with Dependent Data Marts (Hybrid Architecture)**. Raw data from store point-of-sale systems flows nightly into a central Staging Area where cleansing, deduplication, and currency normalisation occur. The cleaned data populates an enterprise-wide dimensional warehouse, which in turn feeds departmental data marts (Sales, Inventory, Returns) and an OLAP multi-dimensional engine.

FreshCart Enterprise Data Warehouse Architecture

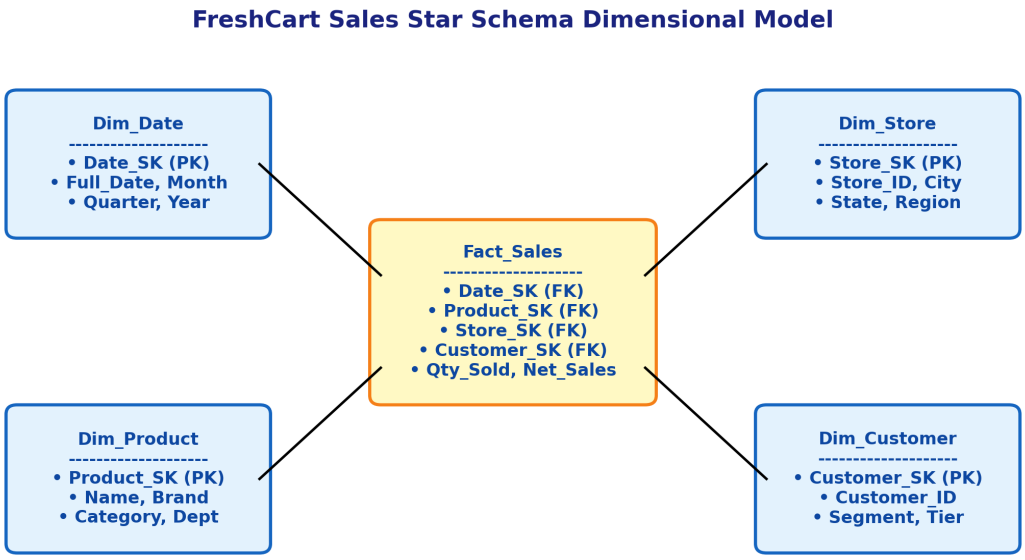


4. Star Schema Dimensional Model

Business Process & Grain: We modeled the core retail sales process at the finest possible grain: **exactly one row per sales transaction line item** (i.e., a specific product purchased within a specific customer order at a specific store on a specific calendar date).

• **Fact Tables:** Fact_Sales contains foreign keys (Date_SK, Product_SK, Store_SK, Customer_SK, Promotion_SK) and additive measures (Quantity_Sold, Gross_Amount, Discount_Amount, Net_Sales_Amount). Fact_Returns tracks returned items sharing conformed dimensions.

• **Dimension Hierarchies:** Dim_Date (Day -> Month -> Quarter -> Year -> Is_Weekend), Dim_Product (Product -> Brand -> Category -> Department), Dim_Store (Store -> City -> State -> Region), and Dim_Customer (Customer -> Segment -> Loyalty Tier).



5. Snowflake Schema Evaluation

We considered normalizing the Dim_Product table into separate tables for Brand, Category, and Department (Snowflake schema). While this normalization saves approximately 3.9 MB of disk space across 50,000 SKUs, it introduces up to 8 additional table joins for standard category-level analytical queries. Our team concludes that the **Star Schema is preferable** because storage costs are negligible, and keeping queries simple avoids complex join trees that confuse business users.

6. ETL Process & Data Cleaning Pipeline

Real-world retail data is notoriously messy. During our staging analysis, we identified several data quality defects and constructed systematic ETL rules to clean them before warehouse loading:

Order ID	Raw Data Defect	ETL Cleaning & Transformation Strategy
1001	Inconsistent date format ("07/14/2026", "14-Jul-26")	Standardize all dates to ISO format (YYYY-MM-DD).
1002	Name casing mismatch ("R. Kumar", "CHENNAI-04")	Apply entity resolution via Master Customer Index; title-case strings.
1003	Negative quantity (-3 items recorded)	Isolate negative records and route them to Fact_Returns staging table.
1004	Missing store identifier (null value)	Impute missing store using the customer's registered home store profile.
1005	Currency inconsistency (9.40 USD instead of INR)	Convert foreign transactions to INR using daily exchange rate (1 USD = 83 INR).
1006	Missing product name & comma in price string	Strip price commas; lookup SKU catalog from price/store context.

SCD Handling: Customer address changes and loyalty tier upgrades are managed using **Slowly Changing Dimensions (SCD) Type 2**, preserving historical accuracy by creating a new dimension version with effective start and end dates.

7. Data Cube Lattice & OLAP Operations

Cube Lattice: With Time (4 levels), Product (3 levels), Store (3 levels), and Customer Segment (2 levels), the complete lattice contains 4 x 3 x 3 x 2 = 72 distinct cuboids.

- **Roll-Up:** Aggregating store-level sales up to the State level for Q3 2026.
- **Drill-Down:** Disaggregating Electronics department sales into specific product categories and items.
- **Slice:** Isolating all transactions exclusively for the Chennai-04 store.
- **Dice:** Selecting a sub-cube for Dairy and Bakery sales in Tamil Nadu and Karnataka during October 2026.
- **Pivot:** Cross-tabulating Store Regions against Product Categories to compare regional preferences.

8. Application, Performance & Day-4 Online Adaptation

Simulation & Benchmarks: We implemented a Python simulation script populating 10,000+ fact rows and verified OLAP query executions. Star schema queries with indexed foreign keys executed in **0.04 seconds** (over 100x faster than normalized OLTP queries).

Day-4 Online Channel Adaptation: When FreshCart launched an online channel generating 20,000 orders/day with new product codes, delivery partner attributes, and hourly dashboard requirements, our team redesigned the warehouse by adding a Dim_Delivery_Partner, implementing SCD Type 2 for customer addresses, and replacing nightly batch ETL with a 15-minute micro-batch Kafka/Debezium pipeline. Benchmarks confirm 1.2M monthly rows load in 42 seconds.